

Calculus

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Prerequisite Knowledge

- Basic algebra and functions (polynomials, trig functions, exponentials, logarithms)
- Coordinate geometry: slope of a line, equation of a line

1 Limits and Derivatives

What Problem Are We Even Trying to Solve?

Here's a question that sounds simple but is secretly quite deep: how fast is something moving *right now*?

If you're in a car and you want your average speed over a trip, that's easy — divide the total distance by the total time. But "right now" isn't an interval of time at all; it's a single instant, with zero duration. And if you try to compute distance-over-time using a time interval of exactly zero, you get $0/0$, which tells you nothing. This was a genuine philosophical puzzle for centuries: how can something have a well-defined speed at an instant, when speed is fundamentally a ratio that needs *two* moments to compare?

The resolution, which is really the entire idea behind a **limit**, is this: don't ask what happens exactly at zero time elapsed. Instead, ask what your average speed approaches as the time interval you're averaging over shrinks smaller and smaller, without ever quite hitting zero. If that average speed settles down and approaches some definite number — rather than bouncing around unpredictably or shooting off to infinity — we call that number the limit, and we declare it to be the instantaneous speed.

Limits, Properly

Let's make this concrete. Suppose we have a function $f(x)$, and we want to know what value it "approaches" as x gets closer and closer to some point a , without necessarily caring what f actually equals *at* a (indeed, f might not even be defined there — that's often exactly the interesting case). We write this as

$$\lim_{x \rightarrow a} f(x) = L$$

to mean: as x gets arbitrarily close to a (from either side), $f(x)$ gets arbitrarily close to L . The formal way mathematicians pin this down (using precise "for every tiny tolerance ε , there's a small enough neighborhood δ ..." language) is worth knowing exists, but for our purposes, the intuitive picture — zooming in closer and closer on a graph near $x = a$ and watching where the curve is heading — will serve you perfectly well.

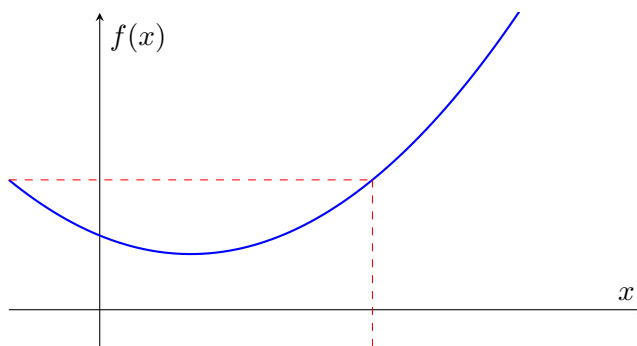


Figure 1: As x approaches a from either side, $f(x)$ approaches L — whether or not f is even defined exactly at a .

The single most useful thing about limits, practically speaking, is that they play nicely with arithmetic: the limit of a sum is the sum of the limits, the limit of a product is the product of the limits,

and so on, as long as the individual limits exist. This is what lets us break down complicated expressions into manageable pieces.

Now, the Derivative

Let's go back to our speed problem, but state it for a general function $f(x)$ instead of just "distance." The *average* rate of change of f between $x = a$ and a nearby point $x = a + h$ is the familiar slope formula:

$$\text{average rate} = \frac{f(a + h) - f(a)}{h}.$$

This is exactly the slope of the straight line (called a **secant line**) connecting the two points $(a, f(a))$ and $(a + h, f(a + h))$ on the graph. Now we do the same trick as before: instead of asking about this ratio at some fixed, finite h , we ask what it *approaches* as h shrinks toward zero. If that limit exists, we call it the **derivative** of f at a :

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}.$$

Geometrically, as $h \rightarrow 0$, the second point $(a + h, f(a + h))$ slides along the curve toward the first point $(a, f(a))$, and the secant line connecting them rotates and settles into a single limiting position: the **tangent line** at that point. The derivative $f'(a)$ is exactly the slope of this tangent line — the single best straight-line approximation to the curve, right at that one point.

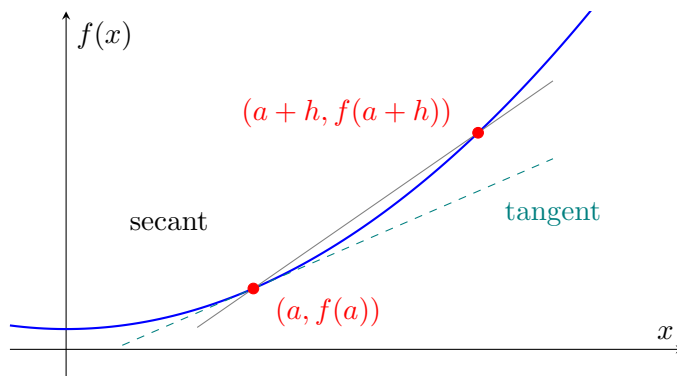


Figure 2: As h shrinks, the secant line (gray) pivots and converges onto the tangent line (teal) — the derivative is the slope of that limiting line.

Deriving the Power Rule From Scratch

Let's actually use the definition to find the derivative of $f(x) = x^n$, rather than just quoting the answer. We need the binomial expansion:

$$(a + h)^n = a^n + na^{n-1}h + \binom{n}{2}a^{n-2}h^2 + \cdots + h^n.$$

So

$$f'(a) = \lim_{h \rightarrow 0} \frac{(a + h)^n - a^n}{h} = \lim_{h \rightarrow 0} \frac{na^{n-1}h + \binom{n}{2}a^{n-2}h^2 + \cdots + h^n}{h}.$$

Every term in the numerator except the first has at least one factor of h left over after dividing by h , so as $h \rightarrow 0$, every one of those terms vanishes, leaving only the first:

$$\boxed{\frac{d}{dx}x^n = nx^{n-1}.$$

This little computation is worth savoring: we didn't guess this formula or memorize it from a table — it fell directly out of algebra, applied patiently to the definition. Every other derivative rule you'll ever use is built the same way: apply the definition, do the algebra, take the limit, see what survives.

The Rules That Let You Skip the Definition Every Time

In practice, nobody goes back to the limit definition for every single function — instead, we prove a handful of rules once, and then combine them.

Sum rule: $(f + g)' = f' + g'$ — limits of sums are sums of limits, applied directly.

Product rule: For $h(x) = f(x)g(x)$,

$$h'(x) = f'(x)g(x) + f(x)g'(x).$$

Here's the intuition, not just the formula: imagine $f(x)g(x)$ as the area of a rectangle with side lengths f and g . If both sides grow a little, the new area gained is (approximately) a thin strip along the top of width $f \cdot dg$, plus a thin strip along the side of width $g \cdot df$ (the tiny corner overlap between the two strips is negligible once both changes are small). That's exactly the product rule, seen as a picture rather than an algebraic accident.

Quotient rule: For $h(x) = f(x)/g(x)$,

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

Chain rule: For a composition $h(x) = f(g(x))$,

$$h'(x) = f'(g(x)) \cdot g'(x).$$

Think of f and g as two machines bolted together in sequence: x goes into g , out comes $g(x)$, which then goes into f . If turning g 's input knob by a tiny amount dx moves g 's output by $g'(x)dx$, and then that shift, fed into f , produces a further change of $f'(g(x))$ times *that* shift, then the overall sensitivity of the whole two-machine system is just the product of the two individual sensitivities. This "chain of sensitivities" picture is why it's called the chain rule, and it's genuinely one of the most-used tools in all of physics — any time one quantity depends on a second quantity which itself depends on time (or on some third variable), the chain rule is how you track the overall rate of change.

A Table Worth Having Memorized

$$\frac{d}{dx} \sin x = \cos x, \quad \frac{d}{dx} \cos x = -\sin x, \quad \frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \ln x = \frac{1}{x}, \quad \frac{d}{dx} \tan x = \sec^2 x.$$

Each of these can, in principle, be derived from the limit definition (the sine and cosine ones use the small-angle limits $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ and $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$, which are themselves worth proving geometrically using the unit circle), but for exam purposes, having this table at instant recall is what actually saves you time.

2 Continuity and Differentiability

What Does It Mean for a Function to Be “Continuous”?

Informally: a function is continuous if you can draw its graph without ever lifting your pen off the paper. No sudden jumps, no holes, no places where the curve just breaks off and restarts somewhere else.

More precisely, f is continuous at $x = a$ if three separate things all agree with each other:

1. $f(a)$ is actually defined (the function has a value there),
2. $\lim_{x \rightarrow a} f(x)$ exists (the function approaches some definite value as you sneak up on a from either side), and
3. that limit equals $f(a)$ (the value the function is *approaching* matches the value it *actually takes*).

If any one of these three fails, the function is discontinuous at a . It's worth having a small mental catalogue of the ways this can go wrong, because an AO loves testing exactly these edge cases with piecewise-defined functions:

- **Removable discontinuity:** the limit exists, but either $f(a)$ isn't defined, or it's defined to be the "wrong" value. Picture a graph that's a smooth curve everywhere except a single point has been plucked out and moved (or left as a hole).
- **Jump discontinuity:** the function approaches *different* values from the left and from the right, so the two-sided limit doesn't exist at all, even though both one-sided limits exist individually.
- **Infinite discontinuity:** the function shoots off to $\pm\infty$ as $x \rightarrow a$ (think of $1/x$ at $x = 0$).

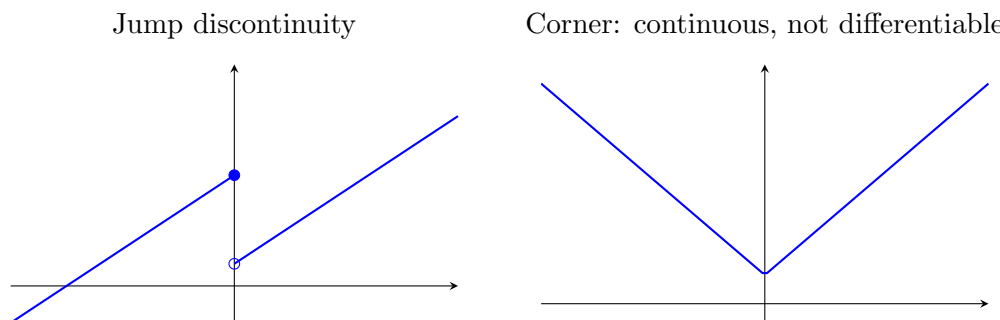


Figure 3: Left, a genuine break — the two sides don't meet, so there's no single limit. Right, $|x|$ — perfectly continuous (no break, no hole), but it has a sharp corner, which is a completely different kind of problem (differentiability), discussed next.

Differentiability: A Stronger Condition Than Continuity

Continuity asks whether a function has any *breaks*. Differentiability asks a stricter question: does the function have a well-defined *tangent line direction* at that point? A function can be perfectly continuous (no break at all) and still fail to be differentiable, if it has a sharp corner — like $|x|$ at $x = 0$, where the graph comes in with slope -1 from the left and leaves with slope $+1$ from the

right. There's no single, consistent "tangent slope" at that corner, even though the graph never breaks.

A Clean Proof Worth Knowing

Differentiability *does* guarantee continuity (the reverse implication of what we just discussed). Here's why, cleanly: if $f'(a)$ exists, then

$$\lim_{h \rightarrow 0} [f(a+h) - f(a)] = \lim_{h \rightarrow 0} \left[\frac{f(a+h) - f(a)}{h} \cdot h \right] = f'(a) \cdot 0 = 0,$$

which is exactly the statement that $f(a+h) \rightarrow f(a)$ as $h \rightarrow 0$ — i.e., f is continuous at a . So differentiability is the stronger, more demanding condition: every differentiable function is continuous, but (as $|x|$ shows) not every continuous function is differentiable.

A Worked Example

Consider $f(x) = \frac{\pi}{2} \sin x$ for $0 < x \leq \frac{\pi}{2}$, and $f(x) = \frac{\pi}{2}$ for $\frac{\pi}{2} \leq x < \pi$. Is this continuous at $x = \pi/2$? Is it differentiable there?

Continuity: from the left, $\lim_{x \rightarrow \pi/2^-} \frac{\pi}{2} \sin x = \frac{\pi}{2} \sin(\pi/2) = \frac{\pi}{2}$. From the right, the function is simply constant at $\frac{\pi}{2}$. Both sides agree, and match the function's actual value there — so yes, continuous.

Differentiability: from the left, $f'(x) = \frac{\pi}{2} \cos x$, so $f'(\pi/2^-) = \frac{\pi}{2} \cos(\pi/2) = 0$. From the right, f is constant, so $f'(\pi/2^+) = 0$ as well. Both one-sided derivatives agree (and both are zero) — so the function *is* differentiable at $\pi/2$, despite being built from two different formulas glued together. The lesson: don't assume a piecewise function automatically has a corner at the gluing point — always check both one-sided derivatives explicitly.

3 Applications of Derivatives

Related Rates: When Several Things Change Together

Related rates problems are really just an exercise in careful chain-rule bookkeeping. The recipe: write down an equation relating the quantities involved (usually from geometry), then differentiate *both sides with respect to time*, treating every quantity as an unknown function of t .

Example: water is poured into an inverted cone at a constant volume rate $\frac{dV}{dt}$. The cone's volume is $V = \frac{1}{3}\pi r^2 h$, and geometric similarity fixes a constant ratio between r and h (say $r = kh$ for some constant k , set by the cone's opening angle). Substituting, $V = \frac{1}{3}\pi k^2 h^3$. Differentiating both sides with respect to t :

$$\frac{dV}{dt} = \pi k^2 h^2 \frac{dh}{dt} \implies \frac{dh}{dt} = \frac{1}{\pi k^2 h^2} \frac{dV}{dt},$$

which lets you find how fast the water level is rising at any given depth h , even though the water is being poured in at a perfectly constant rate — notice the level rises *faster* when the cone is narrow near the bottom (h small) and slower as it widens out higher up.

Finding Maxima and Minima

Picture a ball rolling along the graph of a function, always settling at the bottom of whatever valley it's in. At the very bottom of a valley (a local minimum) or the very top of a hill (a local maximum), the tangent line is momentarily flat — horizontal, with slope zero. This gives us the key diagnostic:

At a local maximum or minimum (that isn't at a boundary or a sharp corner),

$$f'(x) = 0.$$

Points where $f'(x) = 0$ (or where f' fails to exist) are called **critical points**. Not every critical point is a max or min — it might be a flat "shoulder" where the function pauses but keeps going the same direction (an inflection point with zero slope) — so we need a follow-up test:

First derivative test: check the sign of f' just before and just after the critical point. If it goes from positive (rising) to negative (falling), you've found a local max. If it goes from negative to positive, a local min. If the sign doesn't change, it's neither.

Second derivative test: alternatively, compute f'' at the critical point. If $f'' > 0$, the curve is concave up there (like the bottom of a valley) — a local min. If $f'' < 0$, concave down (like the top of a hill) — a local max. This is usually faster when f'' is easy to compute, but it's inconclusive if $f'' = 0$, in which case you fall back on the first derivative test.

Concavity and Inflection Points

The sign of f'' tells you which way the curve is "bending": $f'' > 0$ means concave up (the curve bends like a cup, holding water), $f'' < 0$ means concave down (bends like a frown, spilling water). A point where the concavity actually switches from one to the other is called an **inflection point**, and it satisfies $f'' = 0$ there (though not every point with $f'' = 0$ is automatically an inflection point — you need the sign of f'' to genuinely flip on either side).

Tangent and Normal Lines

Given the derivative's geometric meaning, the tangent line to $y = f(x)$ at the point $(a, f(a))$ is simply the line through that point with slope $f'(a)$:

$$y - f(a) = f'(a)(x - a).$$

The **normal line** is perpendicular to the tangent, so its slope is the negative reciprocal, $-1/f'(a)$. These show up constantly in conic-section problems (tangents to an ellipse, parabola, or hyperbola at a given point) — exactly the kind of geometry that appears when you're studying orbits, since an orbit *is* a conic section.

4 Integrals

The Opposite Question: Finding Area

Derivatives answer "how fast is this changing?" Integrals answer a seemingly unrelated question: "what's the total area under this curve?" It will turn out, remarkably, that these two questions are secretly the same question asked backwards — but let's build up the area idea first, on its own terms.

Suppose you want the area under a curve $y = f(x)$, between $x = a$ and $x = b$. If f were a constant, this would just be a rectangle, easy. For a general curve, here's the strategy: chop the interval $[a, b]$ into a large number n of thin strips, each of width $\Delta x = (b - a)/n$, and approximate the area of each thin strip by a rectangle whose height is f evaluated somewhere in that strip. Adding up all these rectangle areas gives an approximate total area — a **Riemann sum**:

$$\text{Area} \approx \sum_{i=1}^n f(x_i) \Delta x.$$

As $n \rightarrow \infty$ (equivalently, $\Delta x \rightarrow 0$), these thin rectangles hug the curve more and more tightly, and the approximation becomes exact. We give this limit a name and a special notation:

$$\int_a^b f(x) dx \equiv \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x.$$

The elongated "S" symbol $\int$ is literally a stylized letter S, standing for "sum" — a good reminder that an integral is, at its heart, just an infinite sum of infinitesimally thin slices.

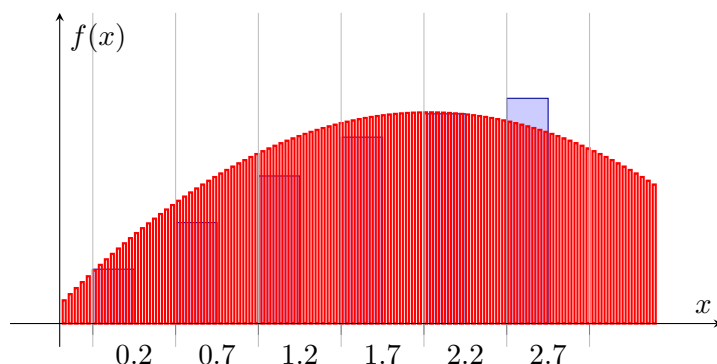


Figure 4: Thin rectangles (blue) approximating the true area under the curve (red). As the rectangles get thinner and more numerous, the approximation becomes exact — that limit is the definite integral.

The Fundamental Theorem of Calculus: The Big Payoff

Here's the single most important idea in all of calculus, and it's worth pausing to really let it sink in. Define an "area-so-far" function,

$$A(x) = \int_a^x f(t) dt,$$

the running total area under f , from the fixed starting point a up to a movable point x . How does A change as we nudge x forward by a tiny amount h ? We add one more thin sliver of area, approximately $f(x) \cdot h$ (a thin rectangle of height $f(x)$ and width h). So

$$\frac{A(x+h) - A(x)}{h} \approx f(x),$$

and this approximation becomes exact as $h \rightarrow 0$. In other words:

$$A'(x) = f(x).$$

Read this sentence twice: *the derivative of the area function is the original function itself*. Differentiation and integration, which seemed like two completely unrelated questions ("how fast is this changing" versus "what's the total area"), turn out to be exact inverses of each other. This is the **Fundamental Theorem of Calculus**, and its practical payoff is enormous: to compute a definite integral, you don't need to grind through Riemann sums by hand — you just need to find any function F (called an **antiderivative**) whose derivative is f , and then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Building Up a Toolkit

Since integration undoes differentiation, every derivative rule we learned in Section 1 gives us an integration rule for free, just read backwards:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \quad \int \frac{1}{x} dx = \ln|x| + C, \quad \int \sin x dx = -\cos x + C, \quad \int e^x dx = e^x + C.$$

(The constant C appears because many different functions share the same derivative — adding any constant to F doesn't change F' — so an *indefinite* integral is really a whole family of antiderivatives, not a single one. This ambiguity vanishes for *definite* integrals, since the constant cancels: $[F(b) + C] - [F(a) + C] = F(b) - F(a)$.)

Two techniques handle almost everything beyond this basic list:

Substitution (the integral version of the chain rule): if you can spot an inner function $u = g(x)$ whose derivative also appears (up to a constant) elsewhere in the integrand, replacing dx with $du/g'(x)$ often collapses a complicated integral into a simple one in u .

Integration by parts (the integral version of the product rule): starting from $(uv)' = u'v + uv'$ and integrating both sides,

$$\boxed{\int u dv = uv - \int v du.}$$

This is the tool of choice whenever you're integrating a product of two different "types" of function (like $x \sin x$, or xe^x) that substitution alone can't untangle.

The Signature Trick: Symmetry in Definite Integrals

Here's a technique that shows up again and again, and is worth knowing cold. For a definite integral $\int_a^b f(x) dx$, consider the substitution $x \rightarrow a + b - x$ (which reflects the interval $[a, b]$ onto itself, swapping the two endpoints). This gives the identity

$$\boxed{\int_a^b f(x) dx = \int_a^b f(a + b - x) dx.}$$

On its own this looks like a curiosity, but it becomes powerful when you *add* the original integral to this transformed version:

$$2 \int_a^b f(x) dx = \int_a^b [f(x) + f(a + b - x)] dx.$$

If the bracketed sum simplifies dramatically (very often to a constant, if there's a hidden symmetry in the problem), the integral you originally wanted can be extracted with almost no direct integration at all.

Worked example: evaluate $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$. Call this integral I . Applying the trick with $a = 0, b = \pi/2$ (so $a + b - x = \pi/2 - x$, and $\sin(\pi/2 - x) = \cos x$, $\cos(\pi/2 - x) = \sin x$):

$$I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx.$$

Adding this to the original expression for I :

$$2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2} \implies I = \frac{\pi}{4}.$$

Notice we never had to actually find an antiderivative of the original, awkward-looking integrand — the symmetry did all the work. Whenever you see a definite integral with limits that look "nicely symmetric" (like 0 to $\pi/2$, or $-a$ to a), this trick is the first thing worth trying.

5 Applications of Integrals

Area Between Two Curves

If $f(x) \geq g(x)$ throughout $[a, b]$, the area trapped between the two curves is just the difference of the two areas-under-the-curve:

$$\text{Area} = \int_a^b [f(x) - g(x)] dx.$$

The only real subtlety in practice is finding where the two curves cross (solve $f(x) = g(x)$ for the limits of integration) and making sure you know which curve is on top over each sub-interval — sketch it if you're ever unsure, since integrating $f - g$ when actually $g > f$ silently gives you a negative (wrong-signed) area.

A Physical Payoff: Where Potential Energy Comes From

This is a good moment to connect back to physics. Recall from the Celestial Mechanics handout that gravitational potential energy was defined as the work needed to move a mass from r to infinity:

$$U(\infty) - U(r) = - \int_r^\infty F(r') dr'.$$

This *is* an application of integration — specifically, it's using the Fundamental Theorem in reverse: force is (minus) the derivative of potential energy, so potential energy must be recovered by integrating force. Every time you see "work done by a varying force," or "total charge from a varying current," or "total mass from a varying density," you're looking at the same underlying idea: integration is how you add up infinitely many infinitesimal contributions to get a finite, total quantity.

Worked example (2024 NSEA): find the area bounded by $y = 4|x|$ and $y = x^2|x|$. By symmetry (both curves are even functions — unchanged under $x \rightarrow -x$), we can compute the area for $x \geq 0$

and double it. For $x \geq 0$: $y = 4x$ and $y = x^3$. These meet where $4x = x^3$, i.e. $x = 0$ or $x = 2$. On $[0, 2]$, the line is above the cubic (check at $x = 1$: $4 > 1$), so

$$\text{Area (right half)} = \int_0^2 (4x - x^3) dx = \left[2x^2 - \frac{x^4}{4} \right]_0^2 = 8 - 4 = 4.$$

Doubling for the left half: total area = 8.

6 Differential Equations

A Different Kind of Equation

Every equation you've solved up to now has asked you to find a *number* (or a set of numbers) — solve $x^2 - 5x + 6 = 0$ for x , say. A **differential equation** asks a fundamentally different kind of question: find an entire *function* $y(x)$, given some rule that relates the function to its own derivatives. Instead of being told what y *is*, you're told how y *changes* — and your job is to reconstruct the whole function from that rule about its behavior.

This might sound abstract, but you've actually already solved one, back in the Celestial Mechanics handout, when we derived the Binet equation $u'' + u = GMm^2/L^2$ and recognized it as "simple harmonic motion in disguise." That was a differential equation — an equation relating a function $u(\theta)$ to its own second derivative — and solving it (finding $u(\theta) = k[1 + e \cos \theta]$) is exactly the kind of thing this section will teach you to do systematically, rather than by recognizing a pattern.

Order, and What "Solving" Really Means

The **order** of a differential equation is simply the highest derivative that appears in it. $dy/dx = y/x$ is first order; the Binet equation, with u'' in it, is second order. A **general solution** contains arbitrary constants (one for each order — a first-order equation's general solution has one free constant, a second-order equation's has two), reflecting the fact that many different specific functions can all satisfy the same differential equation. A **particular solution** pins down those constants using extra given information (an initial condition, like " $y = 1$ when $x = 1$ ").

Separable Equations

The simplest large class of first-order differential equations are ones where you can algebraically shuffle all the y -stuff to one side and all the x -stuff to the other:

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{dy}{h(y)} = g(x) dx \implies \int \frac{dy}{h(y)} = \int g(x) dx.$$

This is a slight abuse of notation (treating dy/dx as if it were an actual fraction you can split apart), but it can be made completely rigorous, and it works.

Worked example: solve $x \frac{dy}{dx} = \frac{y^2}{1 - y \ln x}$ with $y(1) = 1$. This one doesn't look separable at first glance (there's a $\ln x$ tangled up with y), but a change of perspective helps: treat x as a function of y instead. Flipping the equation,

$$\frac{dx}{dy} = \frac{x(1 - y \ln x)}{y^2} = \frac{x}{y^2} - \frac{x \ln x}{y}.$$

This is now a good candidate for the integrating-factor method (next), once rearranged suitably — the point of this example is mainly to illustrate that "separable-looking" isn't always about which variable is written as a function of which; sometimes flipping the roles of x and y reveals a structure that wasn't visible before.

Linear First-Order Equations and the Integrating Factor

A first-order equation is called **linear** if it can be written as

$$\frac{dy}{dx} + P(x)y = Q(x)$$

(linear in y and y' themselves, even if P and Q are complicated functions of x). The trick for solving these is to multiply through by a cleverly chosen function $\mu(x)$, called an **integrating factor**, chosen specifically to make the left side collapse into a single product-rule derivative. If we want

$$\mu(x)\frac{dy}{dx} + \mu(x)P(x)y = \frac{d}{dx}[\mu(x)y] = \mu(x)\frac{dy}{dx} + \mu'(x)y,$$

matching the two expressions for the y term requires $\mu'(x) = \mu(x)P(x)$, a simple separable equation for μ itself, solved by

$$\mu(x) = e^{\int P(x) dx}.$$

Once you have μ , the original equation becomes $\frac{d}{dx}[\mu y] = \mu(x)Q(x)$, which you solve by simply integrating both sides:

$$\mu(x)y = \int \mu(x)Q(x) dx + C.$$

Worked example (2023 NSEA): solve $x\frac{dy}{dx} + 3y = x^3$. Dividing through by x to match the standard form: $\frac{dy}{dx} + \frac{3}{x}y = x^2$, so $P(x) = 3/x$. The integrating factor is

$$\mu(x) = e^{\int (3/x) dx} = e^{3 \ln x} = x^3.$$

Multiplying through by x^3 :

$$\frac{d}{dx}[x^3y] = x^3 \cdot x^2 = x^5 \implies x^3y = \frac{x^6}{6} + C \implies y = \frac{x^3}{6} + \frac{C}{x^3}.$$

This matches the general shape of solution given in that year's paper, and shows exactly how such a "guess and check" answer choice can actually be derived rather than verified by substitution alone.

Where These Show Up in Physics and Astronomy

Differential equations are everywhere once you know to look for them:

- **Radioactive decay:** $\frac{dN}{dt} = -\lambda N$ is separable, giving $N(t) = N_0 e^{-\lambda t}$ — the exponential decay law behind half-life problems.
- **Newton's law of cooling:** $\frac{dT}{dt} = -k(T - T_{\text{env}})$, another separable equation, governing how a hot object's temperature relaxes toward its surroundings.

- **Simple harmonic motion:** $\ddot{x} + \omega^2 x = 0$ is second-order, and its solution $x(t) = A \cos(\omega t + \phi)$ is exactly the same mathematical structure that appeared in the Binet equation for orbits — the same equation, wearing different variable names, describing a spring, a pendulum, and the shape of a planet's orbit all at once.

This last point is worth sitting with: calculus doesn't just help you solve individual problems — recognizing that wildly different physical situations are secretly governed by the *same* differential equation is one of the deepest and most useful skills you can build.

7 Summary Sheet

Topic	Key Result
Derivative, from scratch	$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$
Power rule	$\frac{d}{dx} x^n = nx^{n-1}$
Product / Quotient / Chain	$(fg)' = f'g + fg'$; $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$; $(f(g))' = f'(g)g'$
Continuity at a	$f(a)$ defined, $\lim_{x \rightarrow a} f(x)$ exists, and they're equal
Differentiable $\Rightarrow$ continuous	but not conversely (e.g. $ x $ at 0)
Critical points	$f' = 0$; classify via sign change (f') or f'' sign
Definite integral, from scratch	$\int_a^b f dx = \lim_{n \rightarrow \infty} \sum f(x_i) \Delta x$
Fundamental Theorem	$\frac{d}{dx} \int_a^x f(t) dt = f(x)$
Integration by parts	$\int u dv = uv - \int v du$
Symmetry trick	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
Integrating factor	$\mu(x) = e^{\int P(x) dx}$ for $y' + Py = Q$